Elementary data structures exercises

10-E01 Circular-queue states

Difficulty: Basic

A circular queue of capacity 5 has head=3, tail=1, size=3. Which indices are live? Apply enqueue X, dequeue, and enqueue Y in order. Update all three fields and state whether the queue is full.

10-E02 Sentinel deletion order

Difficulty: Advanced

A doubly linked list has sentinel N with N.next=A, A.next=B, and B.next=N. Give a safe write order for deleting A and prove the meanings of B.prev and N.next afterward.

10-E03 Left-child/right-sibling

Difficulty: Advanced

Root r has children a,b,c; a has children d,e; d has child f. Write the left and sibling representation and explain how to count a's children.

10-E04 Unlink before release

Difficulty: Research

Five slots 0..4 have active list 0→2 and free list 1→3→4, with NIL=−1. Allocate a slot, insert it between 0 and 2, then delete and release 2. List both chains at every step. What happens if 2 is released before it is unlinked from active?

10-E05 A right sibling is not a child

Difficulty: Research

Using the first-child/right-sibling example in the lecture, find the original subtree preorder rooted at a. What order does ordinary binary preorder produce from a if first is treated as binary left and sibling as binary right? Explain the difference.

10-E06 Equal head and tail, two states

Difficulty: Advanced

A capacity-four, size-based circular queue has head=tail=1. Give two legal sizes and state whether the next enqueue and dequeue can run. Compare its capacity with the reserved-slot version.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Elementary data structures exercises (continued)

> This file contains questions only. Complete your derivations before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study